

Program Question_3

```
Implicit none
Integer,Parameter::n=7
Real::x(n),v(n),a(n),t(n)
Integer::i
!Given that Position(x) and Time(t)
x =[30.13, 31.62, 32.87, 33.64, 33.95, 33.81, 33.24]
t =[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6]
!Compute velocity using finite difference
v(1)=(x(2)-x(1))/(t(2)-t(1)) !Forward Difference for first two point
v(n)=(x(n)-x(n-1))/(t(n)-t(n-1))! Backward Difference

Do i=2,n-1
    v(i)=(x(i+1)-x(i-1))/(t(i+1)-t(i-1))! Central Difference formula
End Do

! Compute Accerleration
a(1)=(v(2)-v(1))/(t(2)-t(1)) !Forward Difference for first two point
a(n)=(v(n)-v(n-1))/(t(n)-t(n-1))! Backward Difference

Do i=2,n-1
    a(i)=(v(i+1)-v(i-1))/(t(i+1)-t(i-1))! Central Difference formula
End Do
Write(*,*)" Time(s)    |Velocity    |accerlaeration"
Write(*,*)" _____"
Do i=1,n
    Write(*,'(F6.2,F12.6,F18.6)')t(i),v(i),a(i)
End Do
Write(*,*)"Velocity at t=0 is ",v(1),"cm/s"
Write(*,*)"Velocity at t=0.6 is ",v(n),"cm/s"
Write(*,*)"Accerleration at t=0 is ",a(1),"cm/s"
Write(*,*)"Accerleration at t=0.6 is ",a(n),"cm/s"
```

End Program